



Bicol University
College of Science
Information Technology Department



Digital Archival System for Publication and Knowledge Management Division of Bicol University

Technical Documentation

In Partial Fulfillment of the Requirements for:
WEB SYSTEMS AND TECHNOLOGIES

Submitted to:
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1. Project Overview

Original Project Title:

Digital Archival System for Publication and Knowledge Management Division of Bicol University

Original Authors/Source:

Capstone Project retrieved from the faculty room of Bicol University College of Science Building 2.

Madrona, L., Manalo, S. K., Medallon, L. J., & Daep, P.K. (2023). *Digital archival system for Publication and Knowledge Management Division of Bicol University* [Unpublished Undergraduate Thesis]. Department of Information Technology, Bicol University.

Module Scope:

The paper by Madrona et al. (2023) **Digital Archival System for Publication and Knowledge Management Division of Bicol University** is an undergraduate capstone project that aims to store, organize, and manage Major Final Output (MFO) documents for different colleges under Bicol University. And also to generate reports for the colleges that can be filtered by year, quarter, and document category. This is because the Bicol University Publication and Knowledge Management Division (BU PKMD) uses Google Drive for the collection, management, and preservation of MFO documents, and submission and organization of reports. Security is one of the main issues of the utilization of GDrive, since unauthorized access and modifications are likely to happen.

Rationale:

The paper proposed a system that aids BU PKMD in the efficiency and effectiveness of their paperwork. There are 3 types of users in the system, which have different access levels: Administrator, Research coordinator, and Personnel. And with this, the researchers believe that this centralized digital archival system will facilitate a better workflow, which solves the existing problems and improves the management and security of the documents.

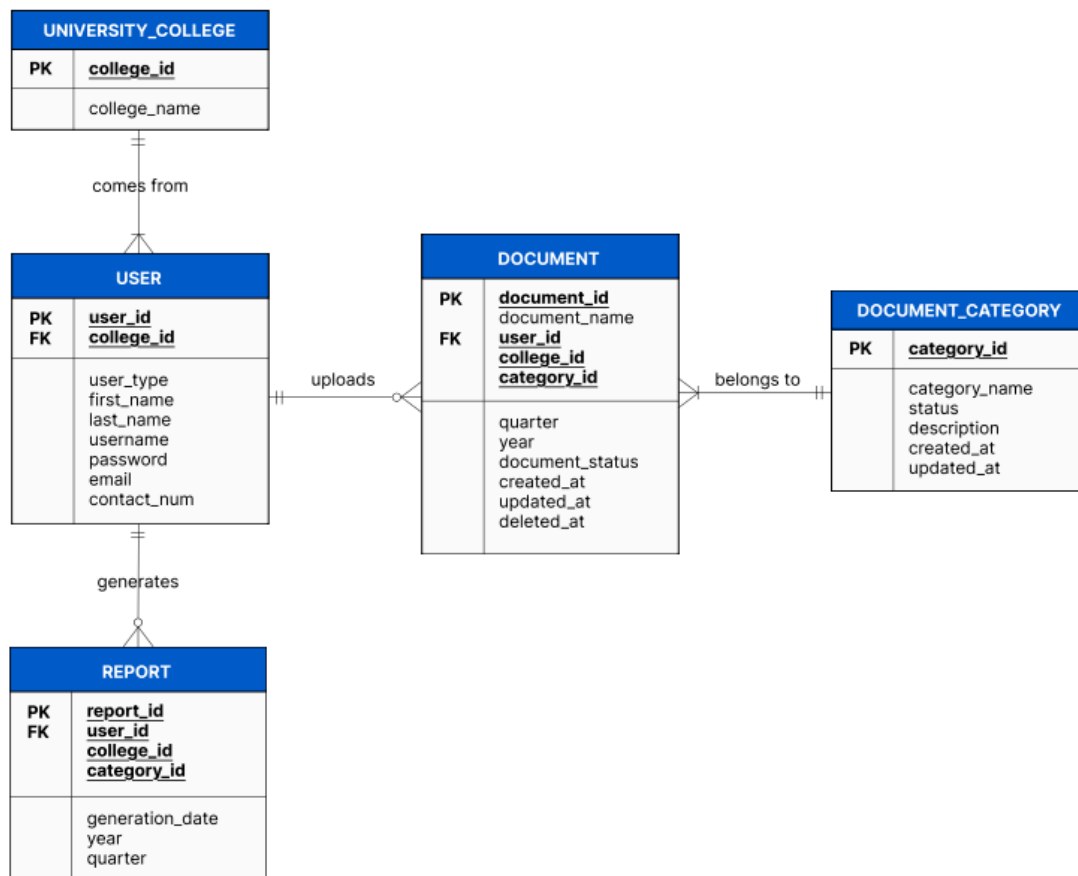
The original implementation utilized PHP, MySQL, XAMPP, HTML, CSS, JavaScript, Chart.js, and Bootstrap 5. PHP for the server-side scripting language paired with HTML, CSS,

and Bootstrap 5 for the frontend. Meanwhile, our implementation uses React to replace PHP and Tailwind CSS to style the frontend. Furthermore, Node.js/Express was used for the backend.

2. Technology Stack

- Frontend Framework: **React** is used for the frontend framework. This JavaScript library allows the website's UI to be dynamic and interactive.
- Styling: **Tailwind CSS** is used for the styling. This utility-first CSS framework is responsible for making the style responsive and consistent directly in the markup.
- Backend Framework: **Node.js/Express** is used for the backend framework. Node.js provides a runtime environment that runs JavaScript on a server, while Express.js is a simple framework that makes managing server-side logic easier.
- Database: **MySQL 8.0** is utilized for the database. This relational database management system (RDBMS) handles the storage of all the data and metadata related to the web application for the original project.

3. Database Design



Relationship	Type	Description
USER → DOCUMENT	One to Many	A user can upload multiple documents to the system.
DOCUMENT → USER	Many to One	Many documents can belong to one user.
USER → UNIVERSITY_COLLEGE	One to One	Each user must belong to one college under the university.
UNIVERSITY_COLLEGE → USER	Many to One	A college under the university can have multiple users
USER → DOCUMENT	One to Many	A user can upload multiple documents.
DOCUMENT → USER	Many to One	Many documents can belong to one user.
DOCUMENT → DOCUMENT_CATEGORY	One to Many	A document category may store many documents.
DOCUMENT_CATEGORY → DOCUMENT	Many to One	Each document can only belong to one category
USER → REPORT	One to Many	A user may generate multiple reports about the document/s
REPORT → USER	Many to One	Multiple reports can be generated by a user.

UNIVERSITY_COLLEGE

Attribute	Data Type	Constraint	Description
<u>college_id</u>	INT	PK, AUTO_INCREMENT	ID of college department
college_name	VARCHAR(100)	NOT NULL	Name of college department

USER

Attribute	Data Type	Constraint	Description
<u>user_id</u>	INT	PK, AUTO_INCREMENT	ID of user
college_id	INT	FK	ID of college department
user_type	ENUM('Administrator', 'Research Coordinator', 'Personnel')	NOT NULL	Roles: Administrator, Research coordinator, or Personnel
first_name	VARCHAR(255)	NOT NULL	First name of user
last_name	VARCHAR(255)	NOT NULL	Last name of user
username	VARCHAR(100)	NOT NULL	Username of user
password	VARCHAR(255)	NOT NULL	Password of user
email	VARCHAR(255)	UNIQUE	User's email address
contact_num	VARCHAR(20)	NOT NULL	User's contact number

DOCUMENT_CATEGORY

Attribute	Data Type	Constraint	Description
<u>category_id</u>	INT	PK, AUTO_INCREMENT	ID of the Category
category_name	VARCHAR(100)	NOT NULL	Name of the Category
status	VARCHAR(50)	NOT NULL	Category Status
description	TEXT	NOT NULL	Brief detail of the category
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Creation date
updated_at	TIMESTAMP	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Modification date

DOCUMENT

Attribute	Data Type	Constraint	Description
<u>document_id</u>	INT	PK, AUTO_INCREMENT	ID of Document
document_name	VARCHAR(255)		Filename of Document
user_id	INT	FK	ID of user
college_id	INT	FK	ID of college department
category_id	INT	FK	ID of Document Category
quarter	ENUM('1st Quarter', '2nd Quarter', '3rd Quarter', '4th Quarter')	NOT NULL	Document quarter
year	YEAR	NOT NULL	Document year
document_status	VARCHAR(50)	NOT NULL	Status of document
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload timestamp
updated_at	TIMESTAMP	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Modification timestamp
deleted_at	TIMESTAMP	TIMESTAMP NULL DEFAULT NULL	Deletion timestamp

REPORT

Attribute	Data Type	Constraint	Description
<u>report_id</u>	INT	PK, AUTO_INCREMENT	ID of report
user_id	INT	FK	ID of user
college_id	INT	FK	ID of college department
category_id	INT	FK	ID of Document Category
generation_date	TIMESTAMP	TIMESTAMP DEFAULT CURRENT_TIMESTAMP	Date of report
year	YEAR	NOT NULL	Year covered of report
quarter	ENUM('1st Quarter', '2nd Quarter', '3rd Quarter', '4th Quarter')	NOT NULL	Quarter covered of report

4. Module Features

Describe each CRUD operation with examples:

Create

This operation allows the uploading of Major Final Output (MFO) documents in PDF, DOC, or DOCX format with a maximum file size of 20MB.

Form Validation Rules:

- Document file can be dragged and dropped or browsed, and the given formats should be either in PDF, DOC, or DOCX with a maximum size of 20MB; files exceeding this size will be rejected.
- Document title is required, and an example placeholder text is given to avoid errors.

- Document Category is required, and a dropdown list is given with the four options: Thesis, Capstone, Research Paper, and Feasibility Study.

Error Handling:

- If any required field is left blank, a prohibited cursor will appear when hovering on the submit button, and it will not work.

Read

This operation fetches and displays the documents in the dashboard with the search by title or ID feature.

Search and Filter Features:

- Users can search documents by title or document ID using the search bar at the top of the table; results update dynamically as the user types

Update

This operation allows the Administrator user type to edit the metadata, such as the category or date of the uploaded document.

Status Update:

- The Administrator can update the status of the document, with 3 dropdown options: Approve, Reject, or Delete.

Delete

- This operation allows the deletion of the uploaded MFO document, with the confirmation feature before executing the deletion from the database.

PROJECT ARTIFACTS

USER: Admin

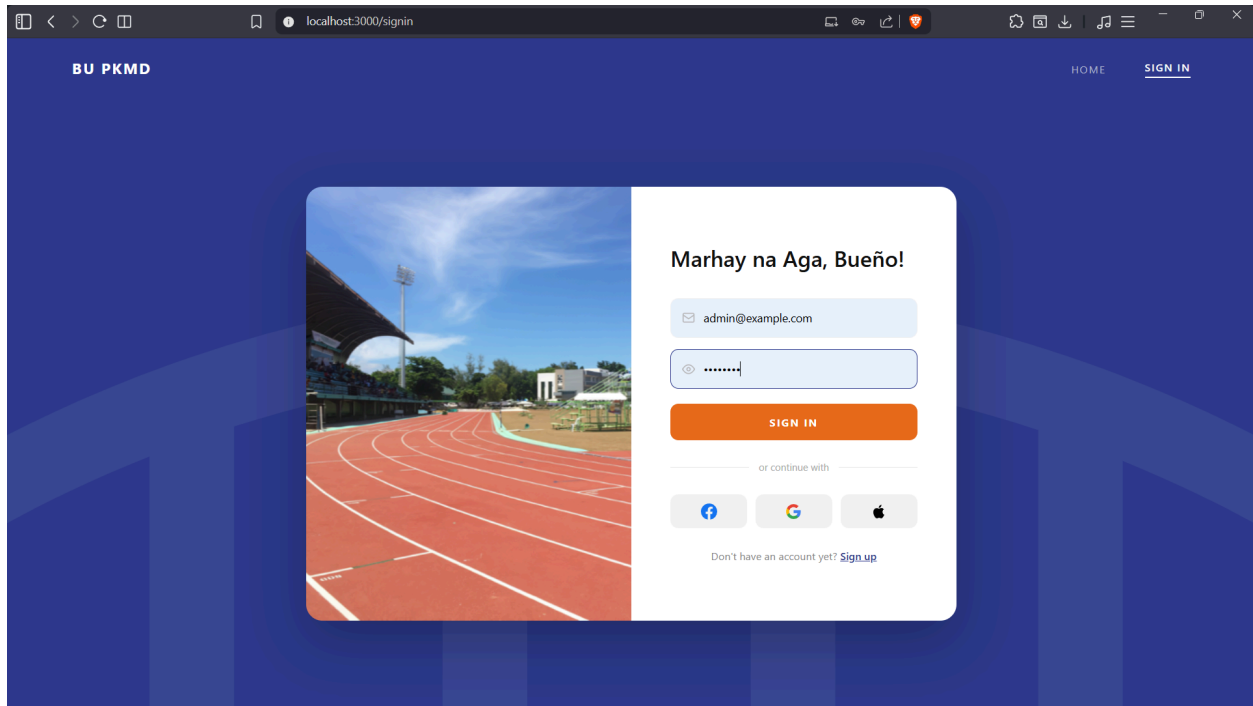


Figure 1. Admin Sign in

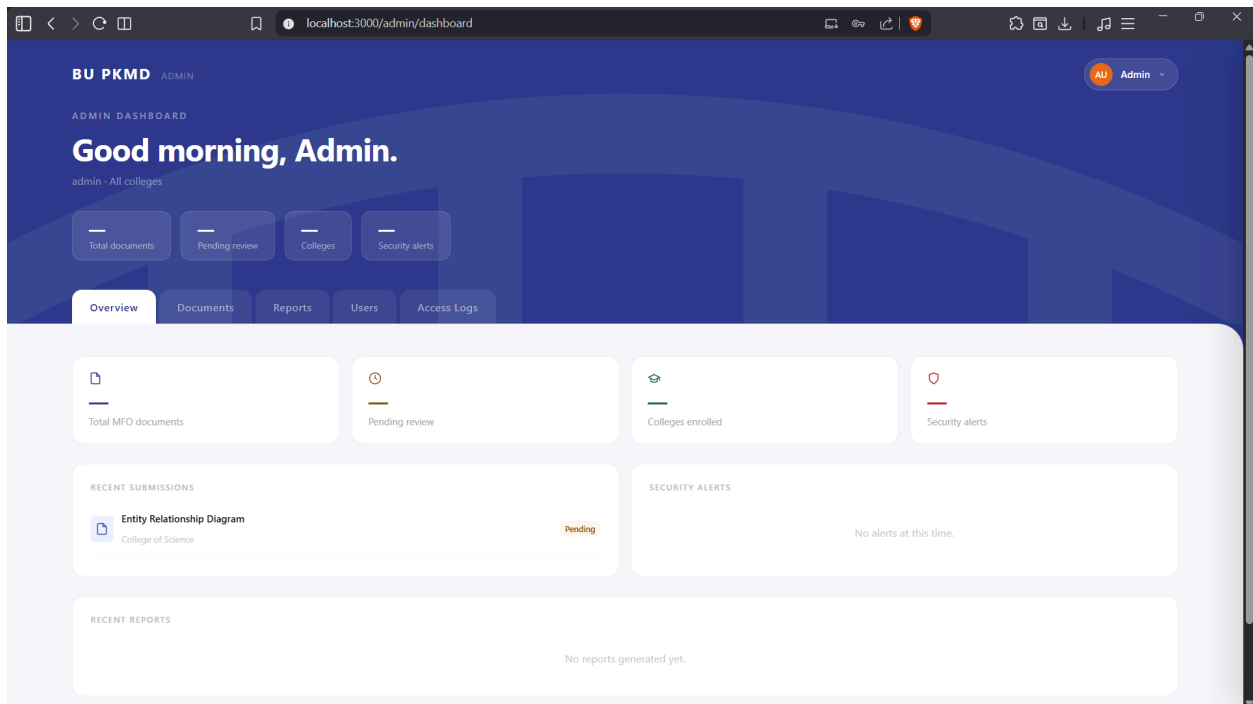


Figure 2. Admin Dashboard Overview

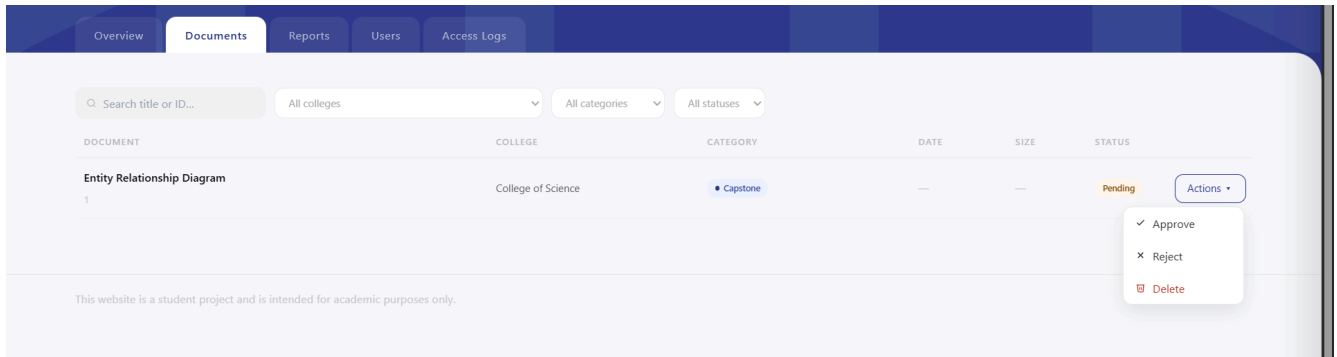


Figure 3. Admin Document Control Dashboard

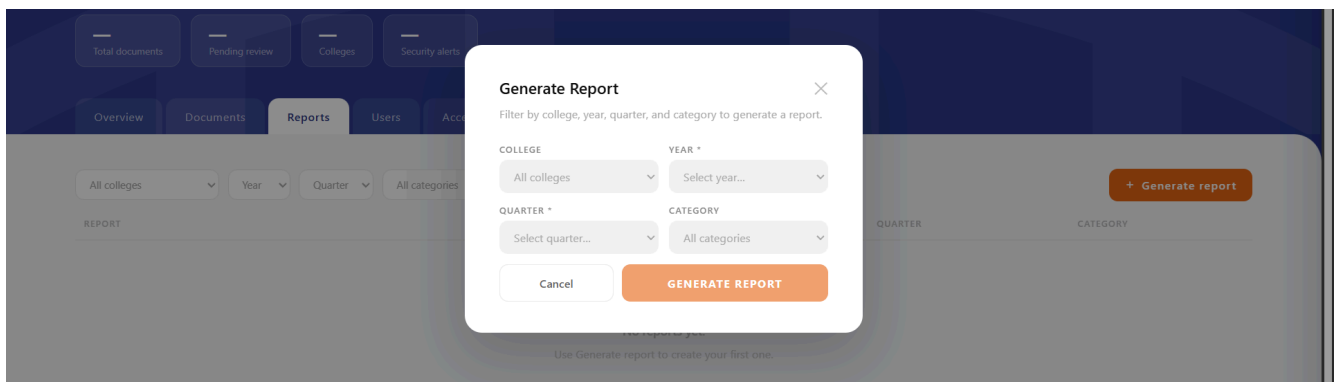


Figure 4. Admin Report Generation Dashboard

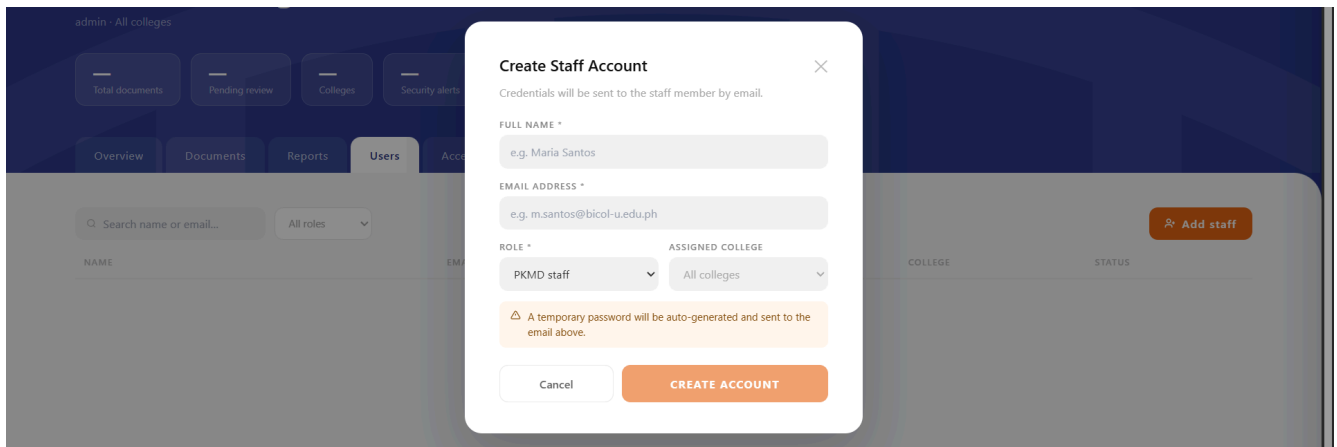


Figure 5. Admin Staff Account Creation Dashboard

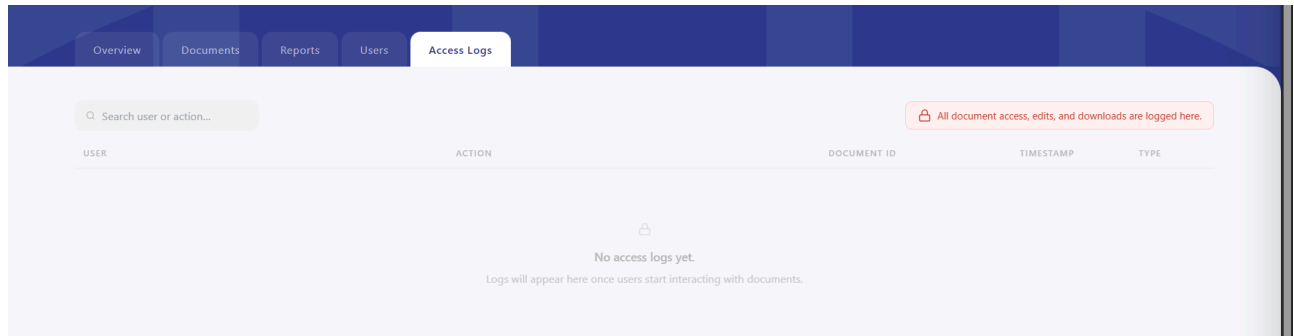


Figure 6. Admin Access Logs Dashboard

USER: Student

Figure 7. Student Account Creation

Create Your Account
Fill in your details to get started and start browsing the library.

COMPLETE NAME *

BU EMAIL ADDRESS *

COLLEGE *

- Select College...
- College of Engineering
- College of Science**
- College of Education
- College of Business, Economics, and Management
- College of Arts and Letters
- College of Social Sciences and Philosophy
- College of Nursing
- Institute of Physical Education, Sports, and Recreation
- College of Law
- Graduate School
- BU Guinobatan
- BU Polangui
- BU Tabaco

DEPARTMENT/COURSE *

PASSWORD *

[Sign in](#)

Figure 7.1 Student Account Creation (Dropdown for College Selection)

Create Your Account
Fill in your details to get started and start browsing the library.

COMPLETE NAME *

BU EMAIL ADDRESS *

COLLEGE *

DEPARTMENT/COURSE *

- Select Department...
- Biology
- Chemistry
- Computer Science
- Information Technology
- Meteorology

YEAR *

PASSWORD *

CREATE ACCOUNT

Already have an account? [Sign in](#)

Figure 7.2 Student Account Creation (Dropdown for Department Selection)

Create Your Account

Fill in your details to get started and start browsing the library.

COMPLETE NAME *	BU EMAIL ADDRESS *
Juan R. Cruz	juan_cruz24@bicol-u.edu.ph
COLLEGE *	DEPARTMENT/COURSE *
College of Science	Information Technology
YEAR *	BLOC *
4th Year	Bloc C
PASSWORD *	CONFIRM PASSWORD *
.....	

CREATE ACCOUNT

Already have an account? [Sign in](#)

Figure 7.3 Student Account Creation (Dropdown for Year and Bloc Selection)

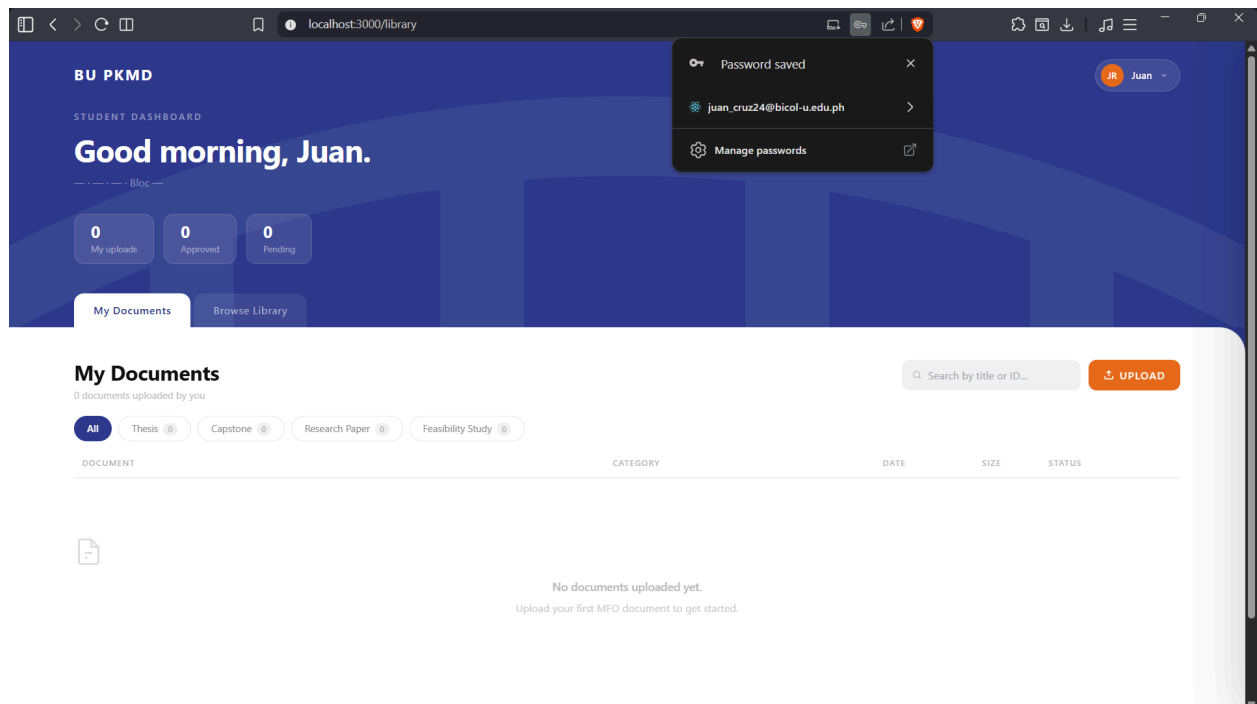


Figure 8. Student Dashboard

Upload MFO Document ✕

Your document will be reviewed by PKMD staff before it appears in the library.



Drag & drop or [browse](#)
PDF, DOC, DOCX · max 20 MB

DOCUMENT TITLE *

e.g. Smart Irrigation System Using IoT

DOCUMENT CATEGORY *

Select category... ▼

[Cancel](#) [SUBMIT DOCUMENT](#)

Upload your first MFO document to get started.

Figure 9. Student MFO document upload

Upload MFO Document ✕

Your document will be reviewed by PKMD staff before it appears in the library.



✓ 1. ERD.pdf
Click to replace

DOCUMENT TITLE *

Entity Relationship Diagram

DOCUMENT CATEGORY *

Capstone ▼

Select category...

Thesis

Capstone

Research Paper

Feasibility Study

Figure 9.1 Student MFO document upload (Document Category Dropdown)

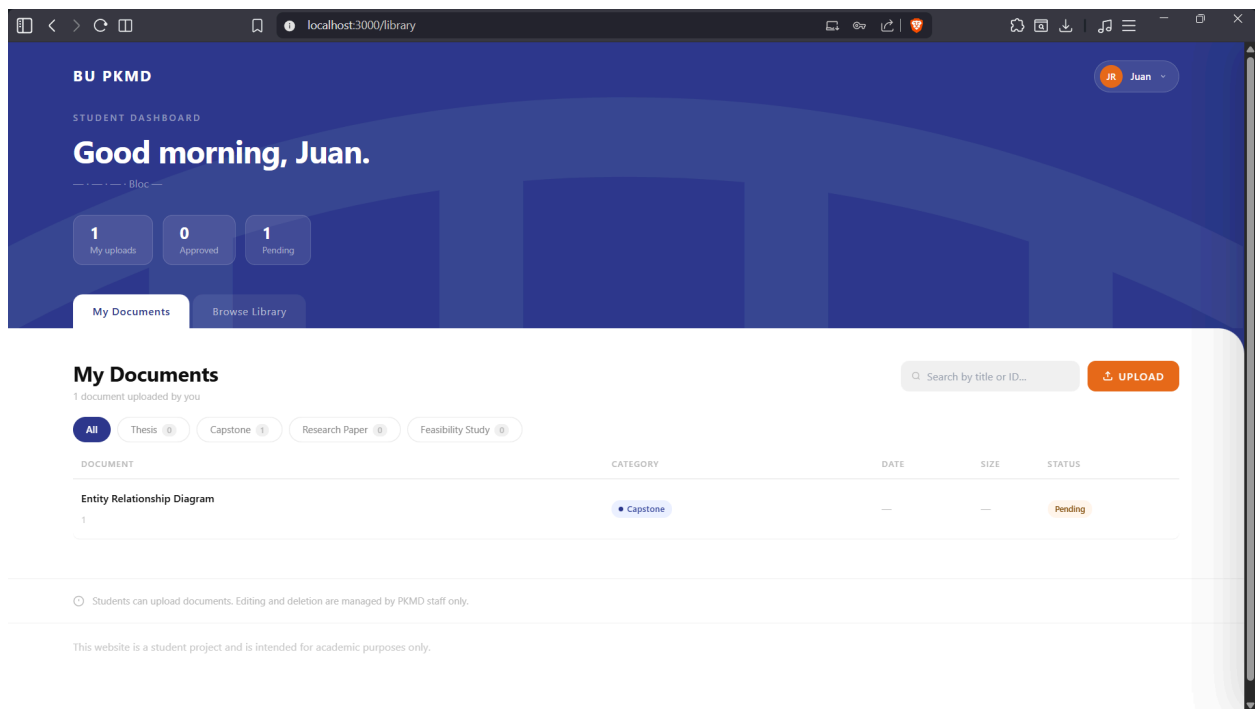


Figure 9.2 Student MFO document upload (Uploaded document with Pending status)

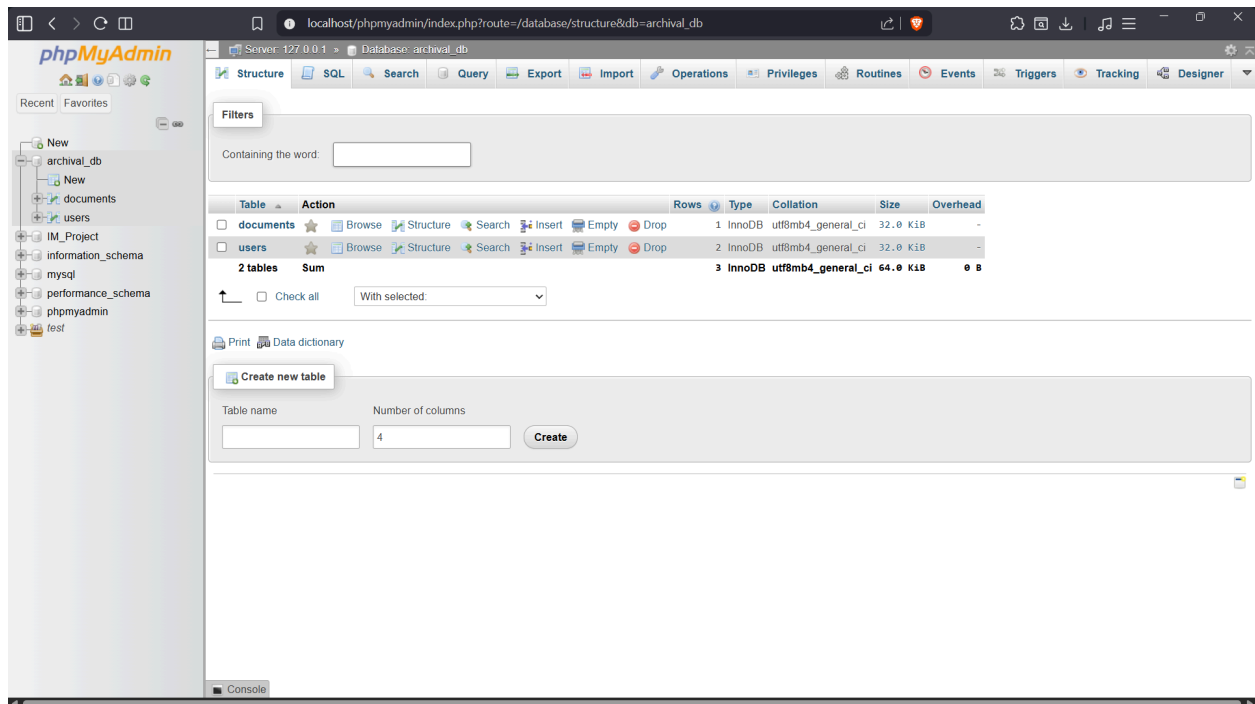


Figure 10. phpMyAdmin database (archival_db)

Showing rows 0 - 1 (2 total, Query took 0.0005 seconds.)

```
SELECT * FROM `users`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	full_name	email	password	college	department	year	bloc	role	created_at
☐	1	Admin User	admin@example.com	\$2b\$10\$tg uSBt80r7vdWSzVWpOT5wM.Bq3V.9ghni3VzOl2...	NULL	NULL	NULL	NULL	admin	2026-05-25 08:39:00
☐	3	Juan R. Cruz	juan_cruz24@bicol-u.edu.ph	\$2b\$10\$CN7IsAYaC4z.WeEX7.d0g.cmEjPoUxyZqJd8UVB67KN...	College of Science	Information Technology	4th Year	C	student	2026-05-25 08:45:15

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Figure 10.1 *phpMyAdmin database (users)*

Showing rows 0 - 0 (1 total, Query took 0.0004 seconds.)

```
SELECT * FROM `documents`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	id	title	category	college	department	file_name	file_path	file_size	status	uploaded_by	uploader_name	upl
☐	1	Entity Relationship Diagram	Capstone	College of Science	Information Technology	1. ERD.pdf	uploads/la7adcc48c7328257789a8bad2d987afc	339 KB	Pending	3	Juan R. Cruz	stuc

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Figure 10.2 *phpMyAdmin database (documents)*